

Homework (Jalapeno)

- Q1.** The function $f(x) = x^2$ is shifted 3 units to the right. What is the new equation?
- (A) $f(x) = (x - 3)^2$
(B) $f(x) = (x + 3)^2$
(C) $f(x) = x^2 + 3$
(D) $f(x) = x^2 - 3$
(E) $f(x) = (x - 3)^2 + 3$
- Q2.** The ancient Egyptians built the Great Pyramid around 2580 BC. If we model the height of the pyramid over time t (in years) using the function $h(t) = -0.1t + 150$, what is the vertical shift of this function?
- (A) 150
(B) 0.1
(C) -0.1
(D) 100
(E) 200
- Q3.** A horizontal reflection over the y -axis is applied to the function $f(x) = x^3$. What is the equation of the reflected function?
- (A) $f(x) = -x^3$
(B) $f(x) = x^3$
(C) $f(x) = (-x)^3$
(D) $f(x) = -(-x)^3$
(E) $f(x) = x^2$
- Q4.** The Rosetta Stone was carved in 196 BC. If the function $p(t) = 2t + 10$ models the number of people p visiting the stone t years after it was carved, what is the y -intercept of this function?
- (A) 10
(B) 20
(C) 5
(D) -10
(E) 0
- Q5.** A vertical stretch by a factor of 2 is applied to the function $f(x) = x^2$. What is the equation of the stretched function?
- (A) $f(x) = 2x^2$
(B) $f(x) = x^2 + 2$
(C) $f(x) = x^2 - 2$
(D) $f(x) = (2x)^2$
(E) $f(x) = (x/2)^2$
- Q6.** The Hanging Gardens of Babylon were one of the Seven Wonders of the Ancient World. If a vertical compression by a factor of 0.5 is applied to the function $f(x) = x^3$, what is the equation of the compressed function?
- (A) $f(x) = 0.5x^3$
(B) $f(x) = x^3 + 0.5$
(C) $f(x) = x^3 - 0.5$
(D) $f(x) = (0.5x)^3$
(E) $f(x) = (x/0.5)^3$
- Q7.** The ancient Greeks used the function $d(t) = 2t$ to model the distance d (in meters) a runner could cover t minutes after the start of a race. What is the horizontal shift of the function $d(t) = 2(t - 5)$?
- (A) 5 minutes
(B) 10 minutes
(C) 2 minutes
(D) 0 minutes
(E) 20 minutes
- Q8.** The Great Wall of China was built to protect the Chinese Empire from invaders. If a reflection over the x -axis is applied to the function $f(x) = -x^2$, what is the equation of the reflected function?
- (A) $f(x) = x^2$
(B) $f(x) = -x^2$
(C) $f(x) = x^2 + 1$
(D) $f(x) = -x^2 + 1$
(E) $f(x) = x^2 - 1$

Open Ended Questions (FRQ)

- Q9.** The function $f(x) = x^2 - 2x + 1$ models the path of a projectile launched from the ancient city of Pompeii. Describe the transformations applied to the parent function $g(x) = x^2$ to obtain $f(x)$, and graph both functions on the same coordinate plane.
- Q10.** Compare the graphs of $y = f(x) = x^3$ and $y = g(x) = -x^3 + 2$. Describe any vertical or horizontal shifts, stretches, compressions, or reflections that have been applied to the graph of $f(x)$ to obtain the graph of $g(x)$. Use complete sentences and provide at least two key points of comparison.

Chef's Tips

- Use historical context to make questions more engaging and relevant to students' interests.
- Ensure that all multiple-choice options are plausible but only one is correct.
- Allow students to use graphing calculators or software to visualize and explore function transformations.